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PRACTICAL 8: VLAN IMPLEMENTATION AND NETWORK SEGMENTATION

Objective

Create VLANs and segment network traffic.

Equipment

- Cisco Catalyst 3560 Switches
- 10 PCs

VLAN Design

VLAN	Department	Devices
VLAN 10	Administration	PC1-PC3
VLAN 20	Lecturers	PC4-PC6
VLAN 30	Students	PC7-PC10

Activities

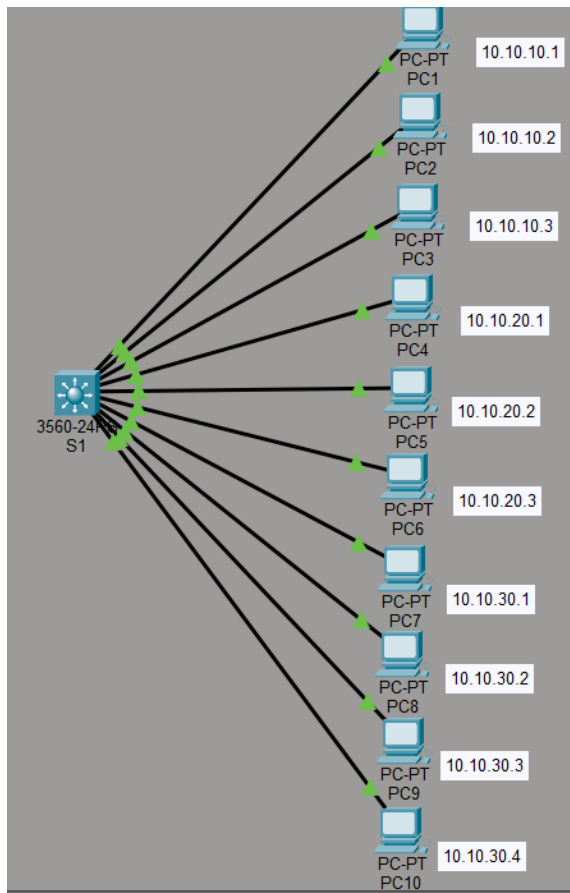


Figure 1: Full topology

1. Create VLANs.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Administration	active	
20	Lecturers	active	
30	Students	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Figure 2: Vlan Config

2. Assign ports to VLANs.

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 Administration	active	Fa0/1, Fa0/2, Fa0/3
20 Lecturers	active	Fa0/4, Fa0/5, Fa0/6
30 Students	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

Figure 3: Vlan membership

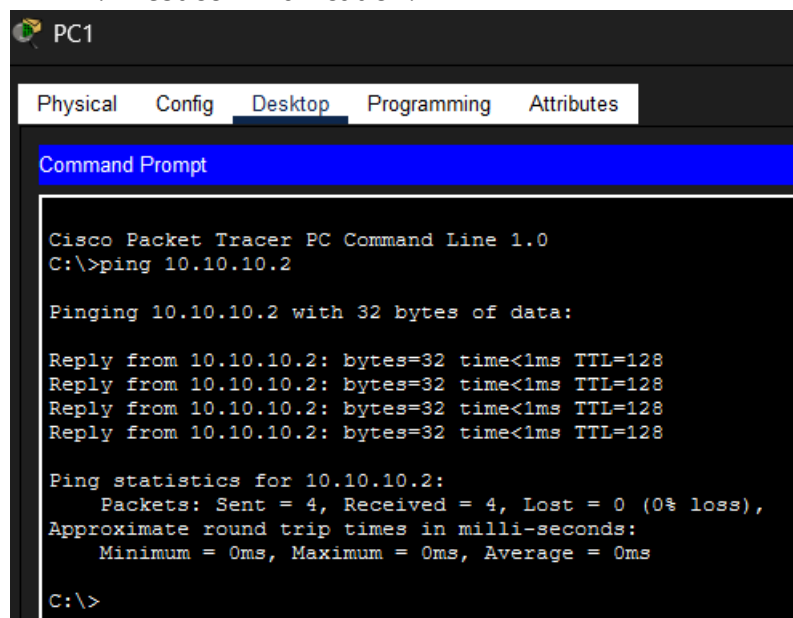
3. Verify VLAN membership.

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 Administration	active	Fa0/1, Fa0/2, Fa0/3
20 Lecturers	active	Fa0/4, Fa0/5, Fa0/6
30 Students	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

4. Test communication.



```
PC1
```

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 4: PC1 to 2

```

C:\>ping 10.10.10.3

Pinging 10.10.10.3 with 32 bytes of data:

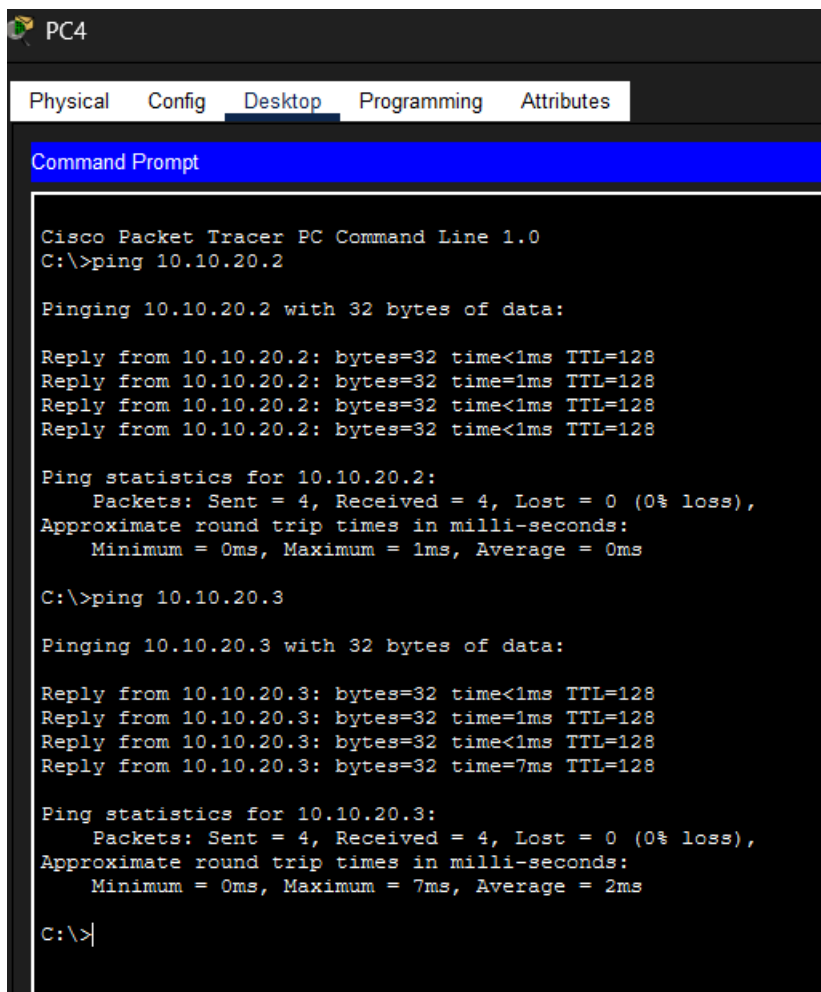
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Figure 5: Ping PC1 to 3



The screenshot shows the 'PC4' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of two ping commands. The first command is 'ping 10.10.20.2', which results in four successful replies with 0ms round trip times. The second command is 'ping 10.10.20.3', which results in four successful replies with round trip times of 0ms, 1ms, 0ms, and 7ms respectively, and an average of 2ms.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.20.2

Pinging 10.10.20.2 with 32 bytes of data:

Reply from 10.10.20.2: bytes=32 time<1ms TTL=128
Reply from 10.10.20.2: bytes=32 time=1ms TTL=128
Reply from 10.10.20.2: bytes=32 time<1ms TTL=128
Reply from 10.10.20.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.20.3

Pinging 10.10.20.3 with 32 bytes of data:

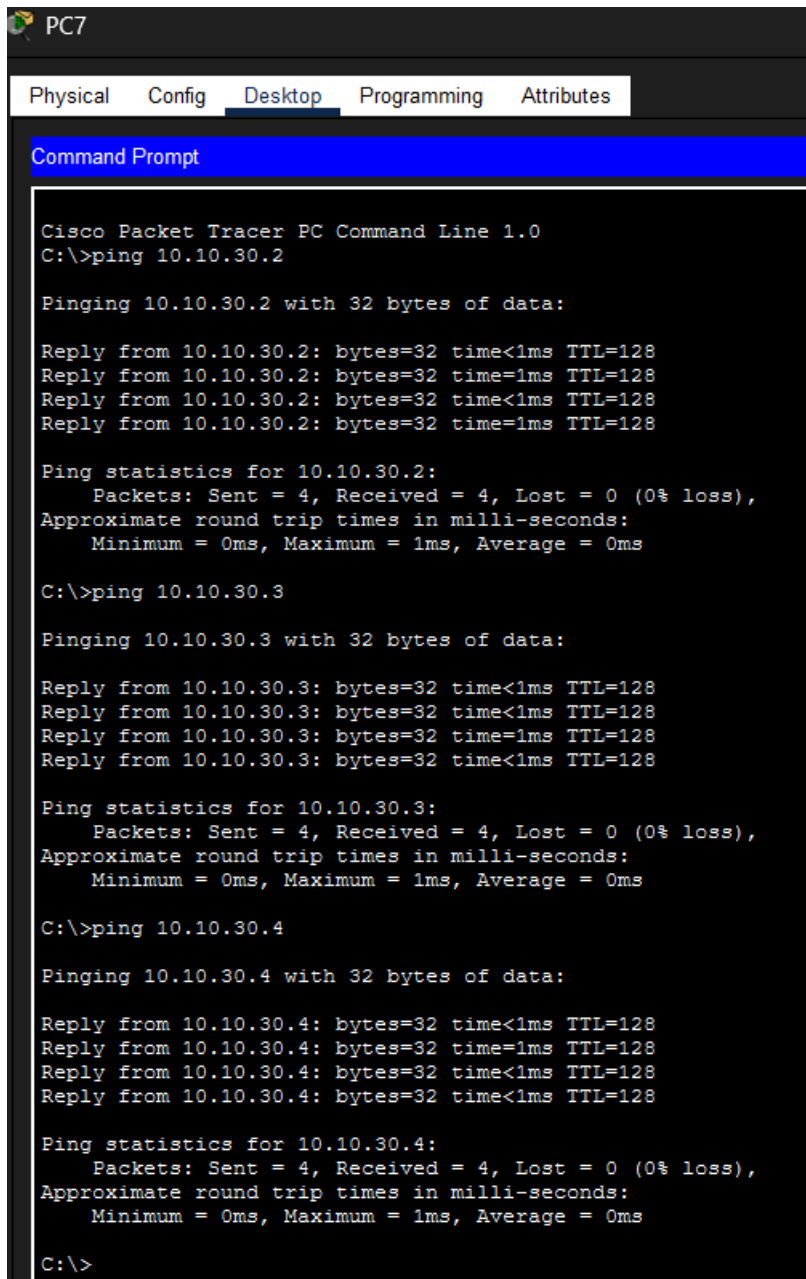
Reply from 10.10.20.3: bytes=32 time<1ms TTL=128
Reply from 10.10.20.3: bytes=32 time=1ms TTL=128
Reply from 10.10.20.3: bytes=32 time<1ms TTL=128
Reply from 10.10.20.3: bytes=32 time=7ms TTL=128

Ping statistics for 10.10.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>

```

Figure 6: VLAN 20 connection



PC7

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.30.2

Pinging 10.10.30.2 with 32 bytes of data:

Reply from 10.10.30.2: bytes=32 time<1ms TTL=128
Reply from 10.10.30.2: bytes=32 time=1ms TTL=128
Reply from 10.10.30.2: bytes=32 time<1ms TTL=128
Reply from 10.10.30.2: bytes=32 time=1ms TTL=128

Ping statistics for 10.10.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.30.3

Pinging 10.10.30.3 with 32 bytes of data:

Reply from 10.10.30.3: bytes=32 time<1ms TTL=128
Reply from 10.10.30.3: bytes=32 time<1ms TTL=128
Reply from 10.10.30.3: bytes=32 time=1ms TTL=128
Reply from 10.10.30.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.30.4

Pinging 10.10.30.4 with 32 bytes of data:

Reply from 10.10.30.4: bytes=32 time<1ms TTL=128
Reply from 10.10.30.4: bytes=32 time=1ms TTL=128
Reply from 10.10.30.4: bytes=32 time<1ms TTL=128
Reply from 10.10.30.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.30.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 7: VLAN 30 connection

```

C:\>ping 10.10.20.1

Pinging 10.10.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.30.1

Pinging 10.10.30.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

Figure 8: VLAN segmentation

5. Document VLAN design.

```

Switch>show vlan brief

```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 Administration	active	Fa0/1, Fa0/2, Fa0/3
20 Lecturers	active	Fa0/4, Fa0/5, Fa0/6
30 Students	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```

Switch>

```

Figure 9: VLAN Verification

Example Commands

```

1  vlan 10
2  name ADMIN
3
4  vlan 20
5  name LECTURERS
6
7  vlan 30
8  name STUDENTS

```

Verification

```

1  show vlan brief

```

Deliverable

Working segmented network.

Assessment Rubric (15 Marks)

Criteria	Marks
VLAN creation	4
Port assignment	4
Connectivity verification	4
Documentation	3
Total	15
